

Freight Rate Prediction

Temporal validation and exploratory analysis

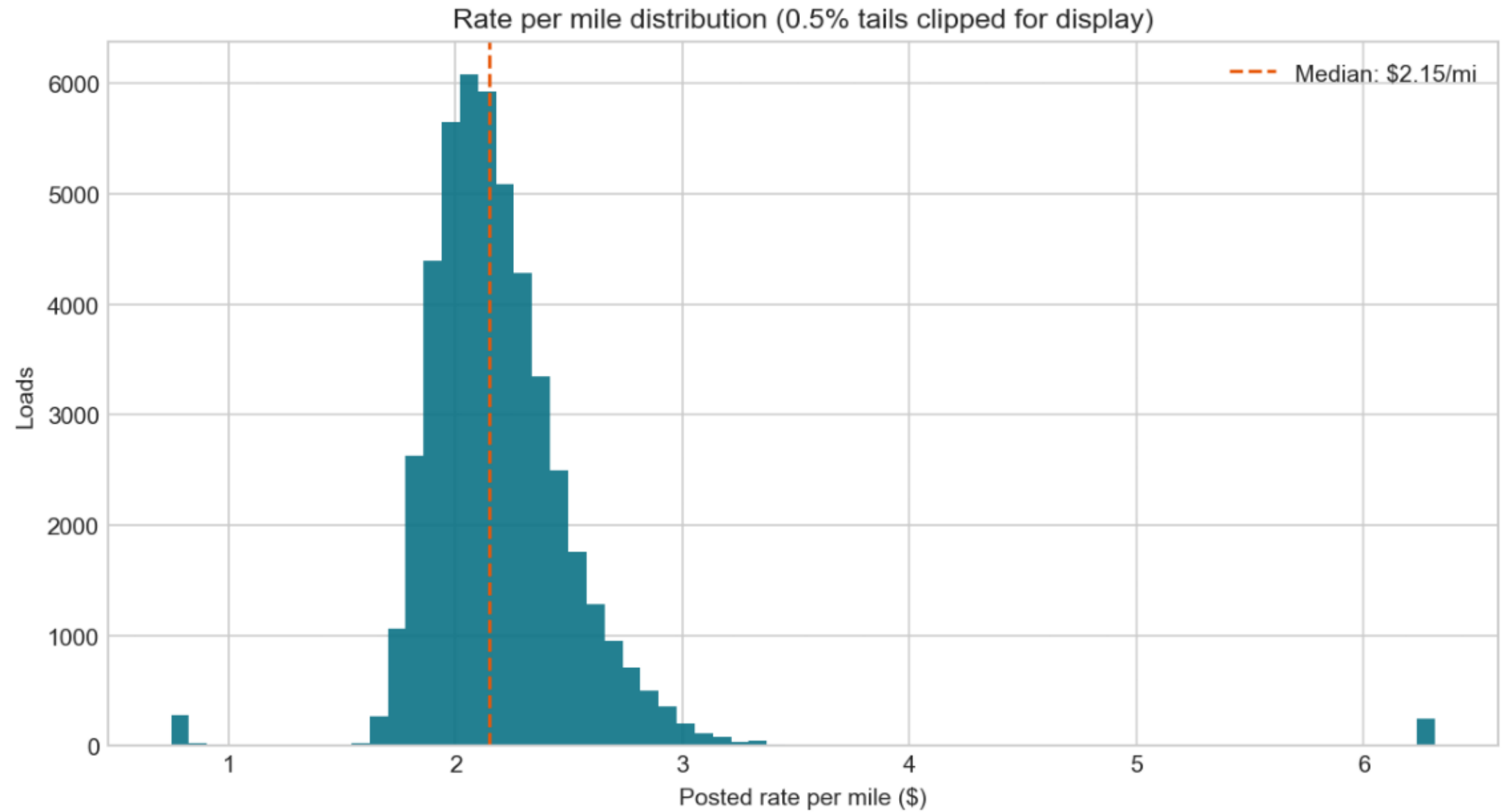
Dataset: 48,000 labeled loads, 2025-01-01 to 2025-10-31
Distance-rate correlation: 0.909
Median rate per mile: \$2.15

Cleaning: absolute weights; equipment-median weight imputation; date/month market-index imputation.
Validation: Jan-Aug training, Sep-Oct temporal holdout.
Model: native-categorical LightGBM trained on `log1p(posted_rate)`.

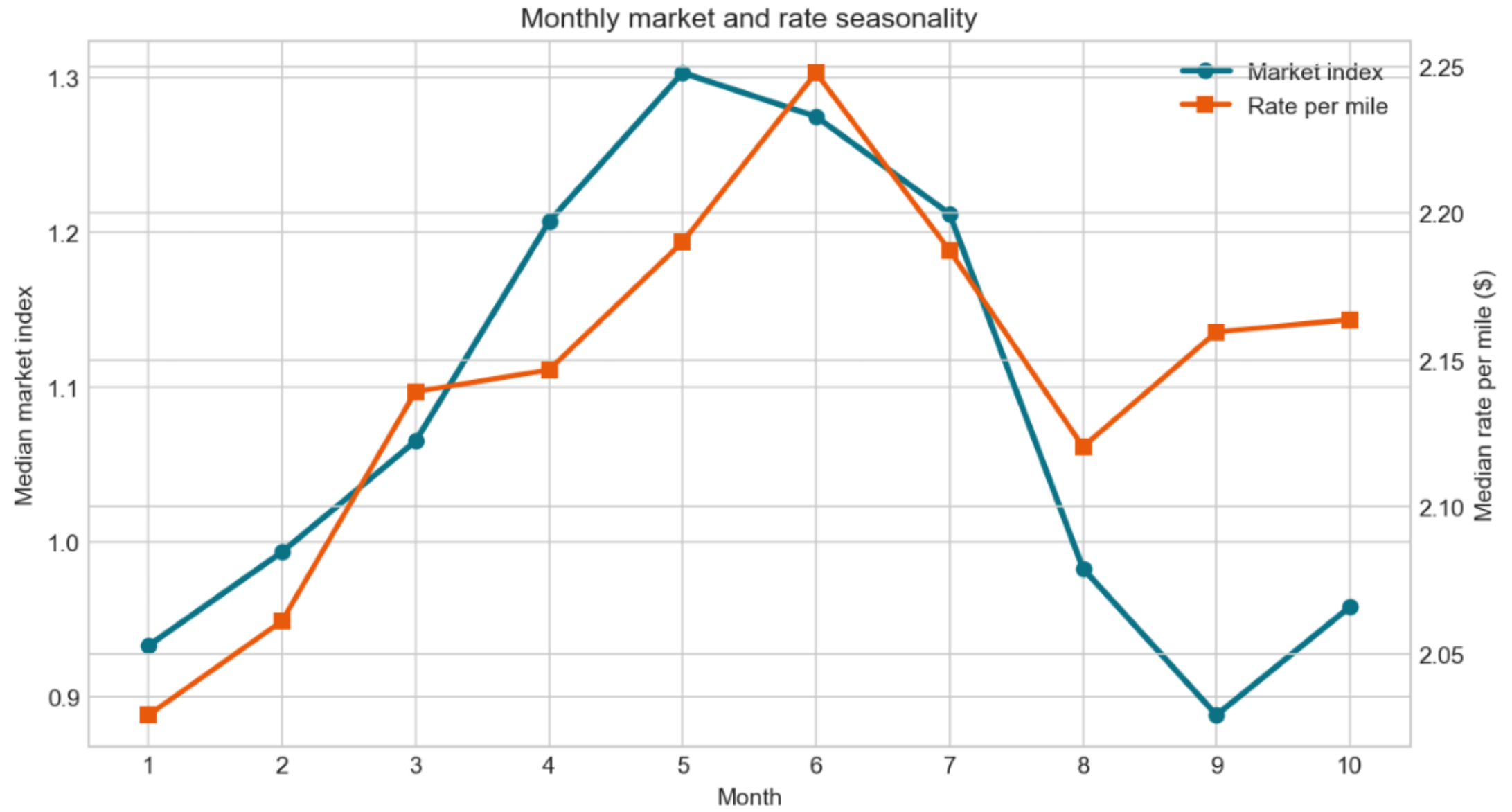
Temporal holdout: train 2025-01-01 through 2025-08-31; holdout 2025-09-01 through 2025-10-31.

	Model	MAE	MAPE	RMSE
Equipment median \$/mi baseline		229.097	10.487	670.660
LightGBM log-target		110.536	4.723	635.282

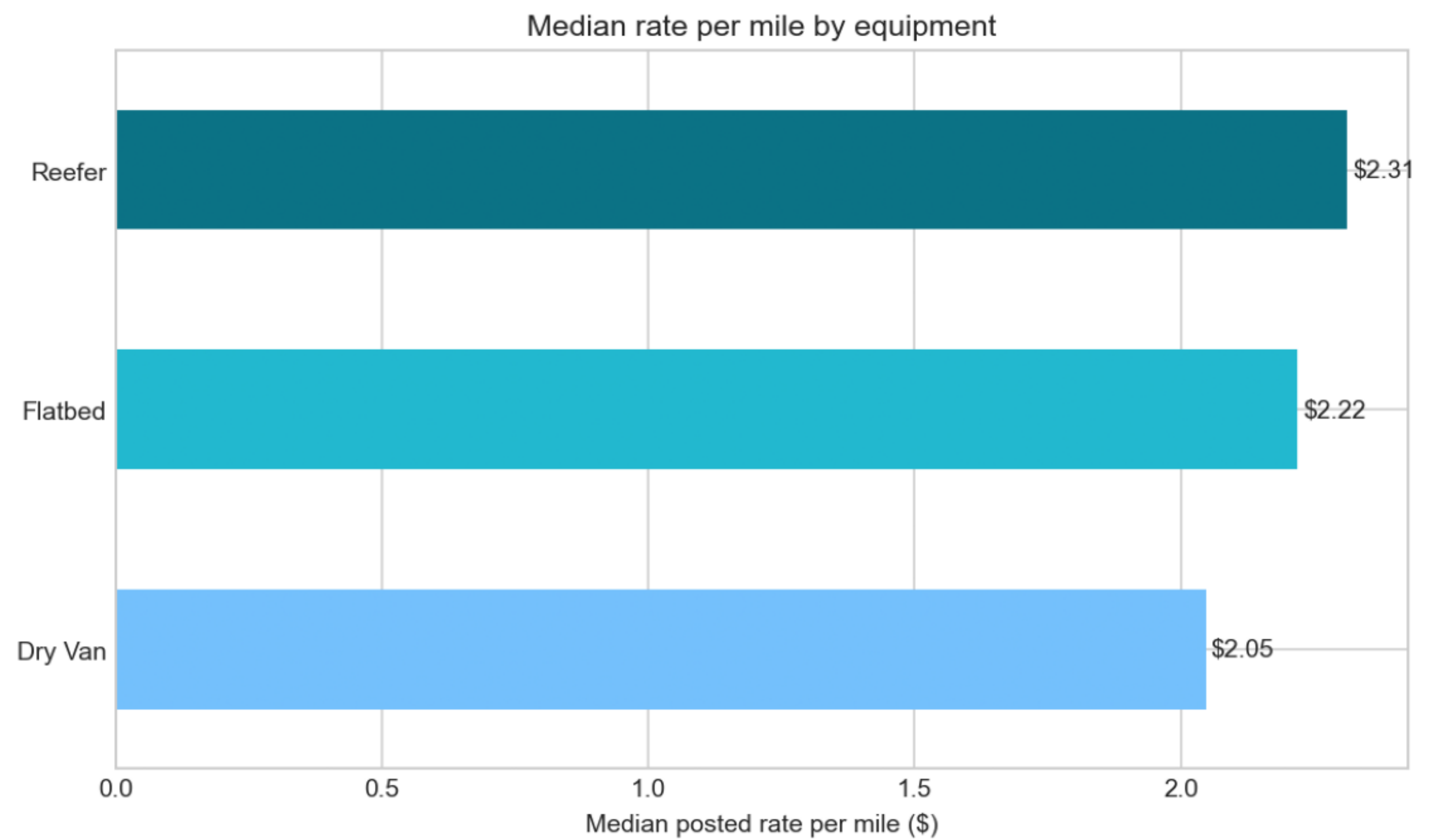
Rate-per-mile distribution



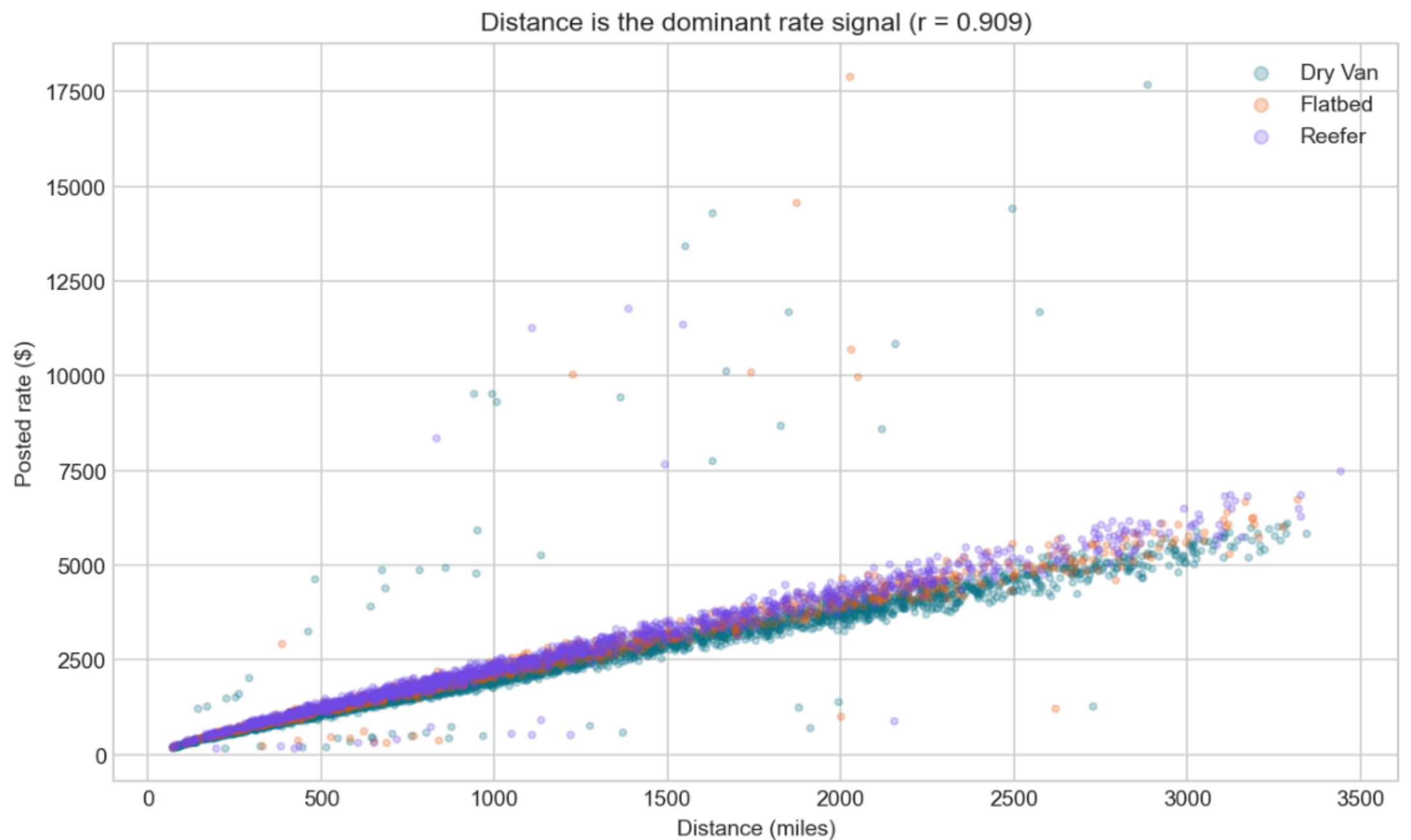
Monthly seasonality



Equipment tiers



Distance versus rate



Model feature importance

